Mohamed Yassine Ben Romdhane

AI & FULL-STACK ENGINEER

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EDUCATION

Engineering Degree in Computer Science — Data Science track

ESPRIM — École Supérieure Privée d'Ingénieurs de Monastir · 2023 – 2026

National engineering degree; final-year project at Capgemini Engineering.

TECHNICAL SKILLS

Languages
TypeScript, JavaScript, Python, SQL

Frontend
React, React Native, Next.js, Angular, Tailwind CSS, Expo

Backend
Node.js, Express, Fastify, FastAPI, Spring Boot, REST APIs

AI & Data
RAG, LLMs, embeddings, pgvector, hybrid retrieval, tool calling, Gemini, NVIDIA NIM, scikit-learn

Databases
PostgreSQL, MongoDB, Redis, Supabase

Tools
Docker, Docker Compose, Git, Socket.IO, MQTT

LANGUAGES

Arabic	Native
French	Fluent
English	Fluent

SUMMARY

AI and full-stack engineer (ESPRIM, 2026) building end-to-end RAG systems and web/mobile platforms. Designed a source-grounded documentation assistant at Capgemini Engineering with FastAPI, pgvector, and multi-provider LLM routing. Builds React and React Native frontends backed by Node.js, Python, and PostgreSQL.

WORK EXPERIENCE

AI Engineer Intern — Capgemini Engineering Feb 2026 – Jun 2026

Final-year project · Source-grounded documentation assistant (RAG)

- Built a Retrieval-Augmented Generation (RAG) assistant that answers natural-language questions across 4 document formats — PDF, DOCX, Markdown, and wiki exports — citing the exact sources behind every answer.
- Designed a backend split into 2 services — Node.js/Express for authentication and conversation orchestration, Python/FastAPI for parsing, chunking, embeddings, and retrieval — with PostgreSQL + pgvector as the shared store.
- Implemented hybrid dense-vector and full-text retrieval with Reciprocal Rank Fusion (RRF), plus a verification pass that rejects unsupported answers.
- Added task-specific LLM routing across 2 providers (NVIDIA NIM for generation and tool calls, Gemini for summaries), query-rewrite retries, and stream-level error handling.
- Validated 11 end-to-end scenarios — access control, source attribution, provider failover, and streaming — all passing, with a median chat latency of 1.9 s on an 18-document corpus in local tests.

Full-Stack & Mobile Engineer — YBInnovations 2025

Project · Wave, a campus community platform for students and associations

- Built a cross-platform Expo/React Native app with 6 modules: events, carpooling, housing, tutoring, deals, and real-time chat.
- Engineered a TypeScript/Express API over PostgreSQL and MongoDB with JWT authentication and per-user data scoping.
- Built a Redis-backed push notification pipeline and integrated Lydia payments with webhook handling.
- Implemented Socket.IO real-time messaging and state management across the 6 modules.

Full-Stack Developer & Data Engineer — MyMatch 2024

Project · Football match booking and management platform

- Developed REST APIs for 3 core workflows: authentication, bookings, and match management.
- Implemented match recommendations from player and match data, with query optimization at the database level.
- Built booking and match-management frontend modules; contributed to schema design and query tuning.